TECHNOLOGY ETHICS • COMPLETE STUDY NOTES

Set 6: Safety and Risk

Safety & Acceptable Risk Criteria, 4 Elements of Risk Management, Risk-Benefit Analysis, Unsafe Design Costs & Strict Liability

PART 6

Safety and risk

2 sections, about 1 h 10 min of reading.

Safety, risk and acceptable risk

~35 min to read | Medium priority | 7 questions

Source: Set 6, pages 4-7 (Unit IV: Safety and Risk). Costs of unsafe design, job risks, the safety-has-a-price curve, knowledge of risk, uncertainties and testing continue in the next section.

Lead-in: engineering as social experimentation and the role of rules

Set 6 opens this part with the closing points of the previous topic. They set the frame for everything on safety.

- Laws and regulations (repeated from t10) also act as a powerful support and defence for engineers who want to act ethically in situations where ethical conduct may not be welcome (for example, a manager who wants a shortcut).
- Viewing engineering as **social experimentation** gives engineers a better perspective on laws and regulations. Every new product is partly an experiment carried out on the public.
- Precise rules and enforceable sanctions are appropriate for misconduct that breaks well-established, regularly re-examined procedures whose purpose is the **safety of the public**.
- In areas of experimentation, rules must not try to cover every possible outcome and must not force the engineer into one rigidly specified course of action. Here regulations should be broad guidelines that still hold the engineer **accountable for his or her decisions**.

The thought experiment and who is affected

Imagine you are a fresh graduate and you are offered a job as an engineer in a large atomic power plant. Would you take it? Under what conditions would you take it, and under what conditions would you refuse? Why? The point of the exercise is that your answer depends on how much risk you judge acceptable, how well you know that risk, and how much control you have over it. Different people will answer differently. That difference is the whole subject of this section.

People as consumers

Group	Meaning	Examples in the slides
Active consumers	Directly involve themselves in using the product, so they have choice and control.	Mowing the lawn, washing clothes, toasting bread.
Passive consumers	Have less choice and less control.	Water, electricity, petrol.
Bystanders	Not users at all, but exposed to the effects.	Exposure to pollution from unknown sources.

What is safe differs by group

- **What is safe to entrepreneurs may not be safe to engineers.** Example: pilots say "Indian airports are not safe; low vision in fog", while the airport owners treat them as acceptable `beyond the slides` (my reading of the example).
- **What is safe to engineers may not be safe to the public.** Example: the top-loading washing machine. The engineer sees it as safe by design standards, but an ordinary user can still be hurt, `beyond the slides` for example by reaching into a spinning drum.

Typically several groups of people are involved in safety matters, and each has its own interests at stake. Each group may differ in what it counts as safe and what it counts as unsafe. So "safe" is never one fixed fact that everyone agrees on.

The concept of safety and why the simple definition fails

The slides give two quotations as the starting definitions.

1. "*A ship in harbor is safe, but that is not what ships are built for.*" (John A. Shedd). A thing that avoids all risk would be useless. Safety cannot mean doing nothing. Engineering products have to be used, and use brings risk.
2. "*A thing is safe if its risks are judged to be acceptable.*" (William W. Lawrence).

Why Lawrence's definition will not do in real life

The slides test it with three cases and ask: how does a judge pass judgement on safety in each?

Case	What goes wrong
We buy an ill-designed iron box in a sale.	**Underestimating risk.** We judge the risk acceptable, but only because we have not seen how dangerous the design is.
We judge that fluoride in water can kill lots of people.	**Overestimating risk.** We reject something that is, in fact, low risk.
We hire a taxi without thinking about its safety.	**Not estimating risk.** We never made a judgement at all.

In all three, "judged to be acceptable" is a bad test, because the judgement is uninformed or distorted. Also, what is acceptable depends on the value judgement of the individual or group. So the definition needs two repairs: the people judging must *know the risks*, and they must judge by their *most settled values*.

The working definition: Martin and Schinzinger

> **Learn this word for word.** "A thing is safe (to a certain degree) with respect to a given person or group at a given time if, were they fully aware of its risks and expressing their most settled values, they would judge those risks to be acceptable (to that certain degree)." (Mike Martin and Roland Schinzinger)

Read it in pieces. *To a certain degree*: safety comes in degrees, not all or nothing. *With respect to a given person or group at a given time*: it is relative to who is judging and when. *Fully aware of its risks*: the judgement must be informed. *Most settled values*: not a panicked or careless mood, but their considered values. *Judge those risks to be acceptable*: the final test is acceptable risk.

Not safe

A thing is **not safe** if it exposes us to **unacceptable danger or hazard**.

> **Q:** **2017 Q5A · 3 marks** **Define the concept of safety as applied in the Engineering profession.**

> **Q:** **CAT 2021-22 Q5a · marks not shown** **Describe your understanding of the safety concept as applied to the Engineer's discipline.**

> **Q:** **CAT 2021-22 Exam Q6A · marks not shown** **Define safety and risk in respect to an engineering design and highlight differences between these two terms.**

MARKER WORDS: THE CONCEPT OF SAFETY

Say it like this:

> A thing is safe **to a certain degree** with respect to a **given person or group at a given time** if, **fully aware of its risks and expressing their most settled values**, they would judge those risks to be **acceptable**. A thing is **not safe** if it exposes us to **unacceptable danger or hazard**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
acceptable risk	giving the one-line answer to "what is safety?"	**MUST**
unacceptable danger or hazard	saying what "not safe" means	**SHOULD**
fully aware of its risks	explaining why the Lawrence definition is not enough	**SHOULD**
most settled values	completing the Martin and Schinzinger definition	**SHOULD**
underestimating, overestimating, not estimating risk	showing why a simple "judged acceptable" test fails	**SHOULD**
given person or group at a given time	explaining that safety differs between groups	**SHOULD**
active consumers, passive consumers, bystanders	listing who is affected by a product	**SHOULD**

▶ **Recall drill: write the definition from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Lawrence's definition of safety.
A thing is safe if its risks are judged to be acceptable.

Martin and Schinzinger's working definition of safety.
A thing is safe (to a certain degree) with respect to a given person or group at a given time if, were they fully aware of its risks and expressing their most settled values, they would judge those risks to be acceptable (to that certain degree).

When is a thing NOT safe?
If it exposes us to unacceptable danger or hazard.

The three ways people mis-judge risk (Set 6).
Underestimating risk; overestimating risk; not estimating risk.

The three types of people as consumers.
Active consumers; passive consumers; bystanders.

Risk, and why risk is reality and safety is fantasy

> **Definition. Risk is the potential that something unwanted and harmful may occur.**

We take a risk when we undertake something or use a product that is not safe. Risk in technology includes the dangers of three kinds of harm:

1. **Bodily harm** (injury, illness, death).
2. **Economic loss**.
3. **Environmental degradation**.

The surprising reversal

Some people assume that safety is a concrete concept and risk is a vague, hypothetical one. The slides say it is the other way round.

- Risks **always exist**.
- True safety **never exists**, except in hypothetical situations.
- So: **risk is reality, safety is fantasy.**

This is why the working definition of safety is built from risk. We cannot remove all risk, so the real question is always how much risk is acceptable.

Q: **2017 Q5B · 4 marks** Explain the term RISK and its relation to safety in Engineering.

Q: **2015 exam Q2a · 2 marks** What do you understand by risk? [reconstructed from garbled scan]

Q: **CAT 2021-22 Q5b · marks not shown** Explain the term risk and how it is related to safety in Engineering.

MARKER WORDS: RISK

Say it like this:

> Risk is **the potential that something unwanted and harmful may occur**: **bodily harm, economic loss or environmental degradation**. **Risk is reality, safety is fantasy**, because risks always exist and true safety never does.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
potential that something unwanted and harmful may occur	defining risk	**SHOULD**
bodily harm, **economic loss**, **environmental degradation**	listing the three kinds of risk in technology	**SHOULD**
risks always exist	explaining why risk is real	**SHOULD**
true safety never exists	explaining why safety is a judgement	**SHOULD**
risk is reality, safety is fantasy	linking risk to safety	**SHOULD**
we take a risk when we use a product that is not safe	relating risk to safety in one line	**SHOULD**

▶ **Recall drill: write the definition from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Define risk.
Risk is the potential that something unwanted and harmful may occur.

The three kinds of danger risk in technology can include.
Bodily harm; economic loss; environmental degradation.

Complete: "Risk is ___, safety is ___."
Risk is reality, safety is fantasy.

When do we take a risk?
When we undertake something or use a product that is not safe.

Safety as acceptable risk, and what makes a risk acceptable

What degree of risk is acceptable? The slides answer in steps.

- Safety is a matter of how people would find risks acceptable or unacceptable **if they knew the risks** and were basing their judgements on their **most settled value perspective**. To this extent, safety is **objective**.
- But perspectives differ between people and groups. To this extent, safety is **subjective**.
- So: **safety is "acceptable risk".**

> **Definition. "A risk is acceptable when those affected are generally no longer (or not) apprehensive about it."**

Apprehension factors

Apprehension means anxiety. Whether people are apprehensive depends largely on these factors:

1. Whether the risk is **assumed voluntarily**.
2. How the **probabilities of harm (or benefit)** are perceived.
3. **Job-related or other pressures** that make people aware of risks or overlook them.
4. Whether the **defects** of a risky activity or situation are **immediately noticeable or close at hand**.
5. Whether the **potential victims are identifiable beforehand**.

A way to remember the pattern: people are calmer when the risk is their own choice, the odds look good, nothing pressures them, the harm is not visible, and nobody in particular is picked out as the victim. They are more anxious when the risk is imposed, visible, close, and aimed at known people.

> **Q:** **CAT 2021-22 Exam Q6C · marks not shown** Under what circumstances would the risk in a given engineering design be considered acceptable and what factors affect this risk acceptability?

MARKER WORDS: ACCEPTABLE RISK AND APPREHENSION

Say it like this:

> Safety is **acceptable risk**: it is **objective** to the extent that people judge informed by the risks, and **subjective** because perspectives differ. A risk is acceptable when **those affected are generally no longer apprehensive about it**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
acceptable risk	concluding what safety is	**MUST**
objective and **subjective**	explaining why safety is partly factual and partly a matter of perspective	**SHOULD**
probabilities of harm	explaining how people perceive risk	**MUST**
apprehensive	defining acceptable risk	**SHOULD**
assumed voluntarily	listing apprehension factors	**SHOULD**
job-related pressures	explaining why people overlook risks	**SHOULD**
defects immediately noticeable; **victims identifiable beforehand**	completing the list of apprehension factors	**SHOULD**

▶ **Recall drill: write the definition from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Define acceptable risk.
A risk is acceptable when those affected are generally no longer (or not) apprehensive about it.

The five factors apprehension depends on.
Whether the risk is assumed voluntarily; how the probabilities of harm (or benefit) are perceived; job-related or other pressures that cause people to be aware of or overlook risks; whether the defects of a risky activity are immediately noticeable or close at hand; whether the potential victims are identifiable beforehand.

What does apprehension mean in Set 6?
Anxiety.

Voluntary risk, control and disaster

A person takes a **voluntary risk** when he is subjected to risk by his own actions or by actions taken by others, and **volunteers to take that risk without any apprehension**.

Example. John and Ann Smith enjoy riding motorcycles over rough ground for amusement. They take voluntary risk, as part of being engaged in a potentially dangerous sport.

Control

Voluntary risk is connected to **control**. The Smiths know the accident figures for the sport are high. Yet, like most people, they show an unrealistic confidence when they believe the dangers are under their control. In terms of engineering as social experimentation, **people are more willing to be the subjects of their own experiments than of someone else's**, whether social experiment or not.

Chauncey Starr

Starr found that individuals are more ready to assume voluntary risks than involuntary risks, **even when the voluntary risks are 1000 times more likely to produce a fatality** than the involuntary ones. So a risk that is imposed (a nearby plant, a passive-consumer service) is judged far more harshly than a risk we choose (a sport, driving), regardless of the numbers. This is why engineers cannot decide acceptability from statistics alone.

Disaster

> **Definition. A disaster = a seriously disruptive event + a state of unpreparedness.**

Example: the Titanic collision with an iceberg at night. The collision alone was an emergency. Fewer lifeboats than needed, inadequate training, and iceberg warnings that were not heeded turned the emergency into a disaster. `beyond the slides` Preparedness is therefore an engineering responsibility: it decides whether a serious event stays an emergency or becomes a disaster.

MARKER WORDS: VOLUNTARY RISK, CONTROL AND DISASTER

Say it like this:

> A person takes **voluntary risk** when he is subjected to risk by his own actions or others' actions and **volunteers to take it without any apprehension**. A **disaster** is **a seriously disruptive event plus a state of unpreparedness**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
voluntary risk	explaining why some risks are accepted more readily	**SHOULD**
control	explaining unrealistic confidence in risky activity	**SHOULD**
more willing to be the subjects of their own experiments	linking voluntary risk to engineering as social experimentation	**SHOULD**
Starr: voluntary risks accepted even when 1000 times more likely to be fatal	showing that numbers alone do not decide acceptability	**SHOULD**
seriously disruptive event + state of unpreparedness	defining a disaster	**SHOULD**

▶ **Recall drill: write the definition from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

When does a person take "voluntary risk"?
When he is subjected to risk by either his own actions or action taken by others and volunteers to take that risk without any apprehension.

Define a disaster.
A seriously disruptive event + a state of unpreparedness.

What did Chauncey Starr find about voluntary and involuntary risk?
Individuals are more ready to assume voluntary risks than involuntary risks, even when voluntary risks are 1000 times more likely to produce a fatality than the involuntary ones.

How information is presented changes how risk is perceived

The way information needed for a decision is presented can greatly influence how risks are perceived. The slides show this with a disaster-management experiment in which the lives of **600 people** are at stake. The only options are given to two groups in two different ways, and each person must choose.

Version 1: stated as lives saved

- **Program A:** 200 people will be saved.
- **Program B:** a 1/3 probability that 600 people will be saved and a 2/3 probability that nobody will be saved.

Response: **72%** chose A and **28%** chose B.

Version 2: stated as lives lost

- **Program A:** 400 people will die.
- **Program B:** a 1/3 probability that nobody will die and a 2/3 probability that 600 people will die.

Response: this time only **22%** chose A and **78%** chose B.

The arithmetic: the two versions are identical

Option	Working	Expected lives saved
Version 1, A	200 saved for certain	200
Version 1, B	(1/3 x 600) + (2/3 x 0) = 200 + 0	200
Version 2, A	400 die for certain, so 600 - 400 = 200 saved	200
Version 2, B	Expected deaths (1/3 x 0) + (2/3 x 600) = 400, so 600 - 400 = 200 saved	200

All four options give an expected 200 lives saved and 400 lives lost. Version 2 A is exactly Version 1 A in different words, and Version 2 B is exactly Version 1 B. Yet the share choosing the sure option fell from 72% to 22%. Only the wording changed.

Conclusions (the slides' two)

1. The option perceived as yielding a **firm gain** will tend to be preferred over options whose gains are perceived as risky or only probable. (Version 1: "200 saved" for certain wins.)
2. The option emphasizing **firm losses** will tend to be avoided in favour of options whose chances of success are perceived as probable. (Version 2: "400 will die" for certain is avoided, so people gamble on B.)

Key point. Equal risks can be judged acceptable or unacceptable depending on whether they are described as gains or as losses. So when engineers or companies communicate risk, how they frame it affects what the public accepts. This links back to acceptable risk: apprehension depends on perception, not only on the facts.

Watch out. In an exam, do not just report 72/28 and 22/78. Say that the expected outcome is the same in both versions (200 saved) and that only the framing differs. That is the point of the example.

MARKER WORDS: EFFECT OF INFORMATION ON RISK ASSESSMENT

Say it like this:

> The **manner in which information is presented** can greatly influence **how risks are perceived**. An option perceived as yielding a **firm gain** is preferred, and an option emphasizing **firm losses** is avoided in favour of those that look **probable**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
manner in which information is presented	stating the effect of information on risk	**SHOULD**
perceived risk	explaining that perception, not only fact, drives judgement	**SHOULD**
firm gain	explaining why 72% chose the sure option	**SHOULD**
firm losses	explaining why only 22% chose the sure option	**SHOULD**
probable or only probable	describing the gamble option	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The two conclusions of the 600-lives experiment.
1. The option perceived as yielding firm gain will tend to be preferred over those from which gains are perceived as risky or only probable. 2. The option emphasizing firm losses will tend to be avoided in favour of those whose chances of success are perceived as probable.

The results of Version 1 and Version 2 of the 600-lives experiment.
Version 1 (lives saved): 72% chose A, 28% chose B. Version 2 (lives lost): 22% chose A, 78% chose B.

Next: the costs of unsafe design, job-related risks, why safety has a price, and the uncertainties in estimating risk.

Costs of unsafe design, uncertainty and testing

~35 min to read | Medium priority | 11 questions

Source: Set 6, pages 7-9 (from "Secondary costs of products"). Earlier pages, up to the information-framing example, are in the safety, risk and acceptable risk section.

Secondary costs of products

A product has a **primary cost**, the cost of producing it. If it is designed unsafely it also carries **secondary costs**, which fall on the producer after the product leaves the factory. The slides say the cost of a product is high if it is designed unsafely, and list five items:

- Returns and warranty expenses: faulty units come back and must be repaired or replaced free.
- Loss of customer goodwill: buyers stop trusting the brand.
- Cost of litigation: court cases, legal fees and compensation after injuries.
- Loss of customers due to injuries in using the product.
- Cost of rework and lost time in attending to design problems: engineers and the factory must stop to fix the design.

The point is that skimping on safety does not save money. The saving at the design stage is paid back, usually with interest, through these secondary costs.

Q: **2017 Q5C · 5 marks** **List any five costs associated with unsafe engineering designs.**

Q: **CAT 2021-22 Q5c · marks as set** **Describe any five costs that come with unsafe engineering equipment and/or design.**

Why manufacturer and buyer must understand the risk

The manufacturer must understand the risk in a product:

- to help reduce secondary costs;
- to know the possible risk for purposes of pricing, disclaimers, legal terms and conditions and so on;
- to know the cost of reducing the risks;
- to take a decision before finalizing the design. After the design is fixed, change is expensive.

The buyer must understand the risk in a product:

- to judge whether he or she wants to take the risks;
- to judge whether "risk versus costs" justifies taking the risk.

Link to voluntary risk: a buyer can only take a risk voluntarily if the risk has been disclosed. A buyer who is not told is exposed to an involuntary risk.

Q: **2017 Q5D · 3 + 2 marks · also CAT 2021-22 Q5e** **Explain why a clear understanding of the risks in a product design is necessary for (i) the product manufacturer (3 marks) and (ii) product buyers (2 marks).**

MARKER WORDS: SECONDARY COSTS AND UNDERSTANDING RISK

Say it like this:

> An unsafely designed product carries **secondary costs**, namely **returns and warranty expenses**, **loss of customer goodwill**, **cost of litigation** and loss of customers, beyond its primary production cost. The manufacturer must understand the risk **before finalizing the design**, and the buyer must understand it to judge whether **risk versus costs** justifies taking it.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
secondary costs	listing the costs of an unsafe design	**SHOULD**
returns and warranty expenses	giving the first cost	**SHOULD**
loss of customer goodwill	describing reputation damage	**SHOULD**
cost of litigation	describing lawsuits after injuries	**SHOULD**
cost of rework	describing design problems found late	**SHOULD**
pricing, disclaimers, legal terms and conditions	saying why the manufacturer needs to know the risk	**SHOULD**
risk versus costs	explaining the buyer's judgement	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The five secondary costs of an unsafely designed product.
Returns and warranty expenses; loss of customer goodwill; cost of litigation; loss of customers due to injuries in using it; cost of rework and lost time in attending to design problems.

Why must the manufacturer understand the risk in a product?
To help reduce secondary costs; to know the possible risk for pricing, disclaimers, legal terms and conditions; to know the cost of reducing the risks; to take a decision before finalizing the design.

Why must the buyer understand the risk in a product?
To judge whether he or she wants to take the risks; to judge whether "risk vs. costs" justifies taking the risk.

Job-related risks

Many workers take risks in their jobs in their stride, for example exposure to asbestos. The slides make these points:

- Exposure to risk on a job is in one sense voluntary, since a worker can always refuse the work or may have control over how the job is done.
- In practice workers generally have no choice. They stick to the only job available to them.
- They are not generally informed about exposure to toxic substances and other dangers that cannot be readily seen, smelt, heard or otherwise sensed.
- Occupational health and safety regulations and unions can have a better say in correcting these situations, but things are still far below expected safety standards.
- Engineers designing work stations must take into account the casual attitude of workers on safety, especially when they are paid on piece rate. A worker paid per item has an incentive to skip safety steps to go faster.

> **Key point.** Job risk is only "voluntary" on paper. A worker without another job, and without information about hidden hazards, is not choosing freely. Compare the voluntary-risk idea in the previous section.

Problems engineers face with the public concept of safety

The slides name two problems:

1. An **optimistic attitude**: people believe that things that are familiar, that have not caused harm before and over which they have some control present no risks.
2. A **serious shock** when an accident kills or maims people in large numbers or harms people we know, even though, statistically, such accidents might occur infrequently.

The first makes the public under-estimate risk, so they ignore warnings. The second makes the public over-react to rare events. Engineers must design for the real risk, while dealing with a public whose perception is skewed in both directions.

Q: **2017 Q5E · 5 marks (also CAT 2021-22 Q5d)** List any five questionable or clearly wrong practices by engineers over the course of their work that would negatively affect public safety associated with their projects.

MARKER WORDS: JOB RISKS AND PUBLIC CONCEPT OF SAFETY

Say it like this:

> Job-related risk is only **voluntary in one sense**, because workers usually **have no choice** and are **not informed** of hidden dangers. Engineers must also deal with the public's **optimistic attitude** to familiar things and its **serious shock** at large accidents, even if they are statistically infrequent.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
job related risks	introducing workplace risk	**SHOULD**
have no choice	arguing job risk is not truly voluntary	**SHOULD**
toxic substances not readily sensed	giving an example of hidden danger (asbestos)	**SHOULD**
piece rate	saying why work-station design must allow for casual attitudes	**SHOULD**
optimistic attitude	describing public under-estimation of risk	**SHOULD**
statistically infrequent	describing public over-reaction to rare accidents	**SHOULD**
cutting corners; **falsifying data**	listing wrong practices by engineers	**MUST**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The two problems engineers face about the public concept of safety.
The optimistic attitude that things that are familiar, that have not caused harm before and over which we have some control present no risks; and the serious shock people feel when an accident kills or maims people in large numbers or harms those we know, even though statistically such accidents might occur infrequently.

"Safety in a commodity comes with a price"

The slides pose this as an "Explain" question. The argument has four steps.

1. **Absolute safety is never possible**, and safety can be improved in an engineering product only with an increase in cost.
2. On the other hand, **unsafe products incur secondary costs** to the producer beyond the primary (production) costs: warranty costs, loss of goodwill, loss of customers, litigation costs, downtime costs in manufacturing and so on.
3. The cost curves. **P** is the primary cost. It is high for a highly safe (low risk) product and low for a risky one. **S** is the secondary cost. It is high for a highly risky (less safe) product and low for a very safe one.
4. Adding them gives the total cost curve **T = P + S**. There is a point **M** at which the total cost is a minimum, and the cost cannot be reduced below it.

If the risk at the minimum total cost point M is *not acceptable*, the producer must choose a lower, acceptable risk value. The total cost is then higher than at M, and the product is designed accordingly. So safety "comes with a price" in the sense that going beyond the cheapest point costs extra money.

The figure, described in words

The original figure is not reproduced in the notes. This is how to draw it. The horizontal axis is *risk*, running from low risk (very safe) on the left to high risk on the right. The vertical axis is *cost*.

- Curve P (primary cost) starts high at the left, where the product is very safe, and falls steadily as risk increases to the right.
- Curve S (secondary cost) starts low at the left and rises steeply towards the right, where the product is risky.
- Curve T = P + S is the sum of the two at each risk level. It is U-shaped: high at both ends, with a lowest point M in the middle, where P is falling and S is rising at equal rates.
- If the risk at M is acceptable, design at M. If it is not, move left along T to a lower risk, and accept a higher total cost.

The numbers below are **illustrative only**, invented to show the shape. They are not from the slides.

Risk level (illustrative)	P: primary cost	S: secondary cost	T = P + S
1 (very low risk)	90	5	95
2	60	15	75
3	40	30	**70 (minimum, point M)**
4	30	50	80
5 (very high risk)	25	85	110

P falls and S rises as risk goes up. T is lowest at level 3. If level 3 is judged an unacceptable risk, the producer must go to level 2 and pay a total of 75 instead of 70. That extra 5 is the price of safety.

Q: **Set 6 slide question · marks as set** "Safety in a commodity comes with a price." Explain.

MARKER WORDS: SAFETY COMES WITH A PRICE

Say it like this:

> **Absolute safety is never possible**, and safety can be improved only with an **increase in cost**, but unsafe products incur **secondary costs**. The **total cost T = P + S** has a **minimum** at point M, and if the risk there is not acceptable the producer must choose a lower risk at a higher total cost.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
absolute safety is never possible	opening the explanation	**SHOULD**
primary costs P	describing the cost of producing a safer product	**SHOULD**
secondary costs S	describing the cost of an unsafe product	**SHOULD**
T = P + S	describing the total cost curve	**SHOULD**
minimum total cost point M	saying where cost cannot be reduced further	**SHOULD**
acceptable risk	saying the producer may need to go below M	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The total cost curve in the "safety comes with a price" argument.
T = P + S, where P is the primary (production) cost and S is the secondary cost. T has a minimum at point M.

What does the curve P show?
Primary costs are high for a highly safe (low risk) product.

What does the curve S show?
Secondary costs are high for a highly risky (less safe) product.

Knowledge of risk for better safety

Robert Stephenson, the nineteenth-century engineer, wrote that all accidents, the harms caused and the means used to repair the damage should be recorded for the benefit of the younger members of the profession. A faithful account of accidents and of how the damage was contained is more valuable than a description of successful work.

So it is imperative that knowledge of risks will help to attain better safety. But gaps remain, for two reasons:

1. some industries do not freely share information;
2. there are always new applications of old technology that make the available information less useful.

Uncertainties encountered in the design process

The slides list seven sources of uncertainty:

- A decision on maximising profit or maximising the return on investment. This pushes safety margins down.
- Uncertainties about applications, such as dynamic loading instead of static loading, vibrations and wind speeds.
- Uncertainties regarding the materials and skills required in manufacturing.
- Changing economic realities.
- Unfamiliar environmental conditions, such as very low temperature.
- Standard data on items like steel, resistors, insulators and optical glass are based on statistical averages only. A given batch can be weaker than average.
- By the nature of processes, all components have a tolerance in design, so the capability of assemblies is assessed by probability statistics.

Testing strategies for safety

Commonly used testing methods:

- Using past experience in checking the design and performance.
- Prototype testing. The one product tested may not be representative of the population of products.
- Tests simulated under approximately actual conditions, to find performance flaws on safety.
- Routine quality assurance tests on production runs.

These procedures are not always carried out properly, so **testing procedures cannot be trusted uncritically**. Some tests are also destructive, and it is obviously impossible to do destructive testing on every product and still improve safety. In such cases a **simulation that traces hypothetical risky outcomes** can be applied. The slides name two such techniques:

- **Scenario analysis**: start from an event and trace its consequences (event to consequences).
- **Failure modes and effects analysis (FMEA)**: go through the failure modes of each component and work out their effects.

Watch out. Do not mix the two. Scenario analysis begins with an event and asks "what follows?". FMEA begins with a component and asks "how can it fail, and what then?".

MARKER WORDS: KNOWLEDGE OF RISK, UNCERTAINTY AND TESTING

Say it like this:

> Safety is helped by **recorded knowledge of accidents** (Stephenson), but design remains **uncertain** because standard data are **statistical averages** and components have **tolerances**. Testing by **prototype testing**, **simulated tests** and **quality assurance** cannot be trusted uncritically, so for destructive tests engineers use **scenario analysis** and **failure modes and effects analysis**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
Stephenson: accidents should be recorded	explaining knowledge of risk	SHOULD
statistical averages	explaining uncertainty in material data	SHOULD
tolerance	explaining uncertainty in components	MUST
dynamic loading instead of static loading	giving an application uncertainty	SHOULD
prototype testing	listing testing methods	MUST
simulation	handling destructive tests	MUST
failure modes and effects analysis	naming a risk-tracing technique	SHOULD
scenario analysis (event to consequences)	naming the other technique	SHOULD

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The seven uncertainties in the design process.
Maximising profit or return on investment; uncertain applications (dynamic loading, vibrations, wind speeds); uncertain materials and manufacturing skills; changing economic realities; unfamiliar environmental conditions; standard data based on statistical averages only; component tolerance, assessed by probability statistics.

The four commonly used testing methods for safety.
Using past experience in checking design and performance; prototype testing; tests simulated under approximately actual conditions; routine quality assurance tests on production runs.

Two gaps that remain in knowledge of risk.
Some industries do not freely share information; there are always new applications of old technology that render available information less useful.

Scenario analysis.
A simulation tracing hypothetical risky outcomes: Event to Consequences.

Failure Modes and Effects Analysis.
A simulation tracing hypothetical risky outcomes by examining the failure modes of each component.

Practice: uncertainties, testing and the safety questions

Q: **Practice · 8 marks** **Describe the uncertainties an engineer meets in the design process and the testing strategies used to deal with them.**

Q: **Exam 2021-22 Q6B · 4 strategies** **Describe four strategies that should be adopted to ensure that a given engineering design (product) meets the necessary safety requirements.**

Q: **Exam 2021-22 Q6D · 3 reasons** **Give three reasons that necessitate carrying out a risk-benefit analysis of an engineering design.**

Q: **QB Sec B Q2b · 14 marks** **If engineers are not afforded the luxury of seeking perfection, it follows that they must be willing to risk failure. Do you agree? Discuss this in view of engineering product safety and ethics. [reconstructed from garbled scan]**

Risk management process and risk control options beyond the slides

beyond the slides The slides do not cover this, but a past paper asks for it. The standard answer, from general engineering practice, is below.

- Four elements of the risk management process: **risk identification** (find the hazards), **risk analysis** (estimate likelihood and consequence), **risk evaluation** (decide whether the risk is acceptable) and **risk control** (act on it and monitor). Some textbooks use "assessment", "treatment" or "monitoring and review" for the last steps.
- Risk control options: **avoid** the risk (drop the hazardous design or activity); **reduce** it (redesign, add safety devices, train, inspect); **transfer** it (insurance, contracts); **accept** what is left and monitor it.

Q: **2015 Q2b, c · 2 + 6 marks · also QB Sec B Q2a, two papers · beyond the slides** **(b) Identify the four major elements of the risk management process. (c) Engineers seldom manage to eliminate risk in their projects. Identify and explain three risk control options available to an engineer. [reconstructed from garbled scan]**

Q: **2015 Q2d(i) · 2 marks · beyond the slides** **What do you understand by Strict Liability?**